On the prediction of critical transitions



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Abstract

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Chapter 1

Introduction

The dynamics of complex systems has receveid significant attention, over the last decades, from many research communities in science and engineering. Of particular interest to scientists and modelers is the study of the nonlinear response of these systems to external perturbations which often features the emergence of a rich and diverse set of phenomena including pattern formation, chaotic regimes, resonance and, crucially, critical transitions. A critical transition in a complex system, often referred to as a *tipping point* or a *tipping event*, is recorded whenever a small change in the system's input leads to a large and sometimes irreversible change in the systems' response. More specifically, a tipping point is considered as an abrupt shift in the steady–state of a dynamical system following a slow degradation of the stability of its equilibria.

A concrete example, often discussed in the literaure to introduce the main concepts of critical transitions, is the collapse of multi–species ecosystems under sustained and increasing deterioration of their biomass. In 1977 Robert May proposed a dynamical model in which a population of herbivores, sustained by a decreasing level of vegetation cover in a closed environment, can experience a sudden collapse due to unsustainable increases in the grazing rate [53]. In his model, May realized that a characteristic separation between the timescales of the evolution of the animal population and the grazing rate of the biomass leads to qualitatively different regimes in the observed state of the ecosystem. In one state the ecosystem is in a healthy, sustainable equilibrium and the vegetation cover allows for the slowly increase in the grazing activity of the herbivores. Past a certain threshold however (described as *breakpoint* in May's paper) the ecosystem can no longer sustain the animal population and the biomass sharply declines to near–zero across, resulting in the collapse of the herbivores density. This

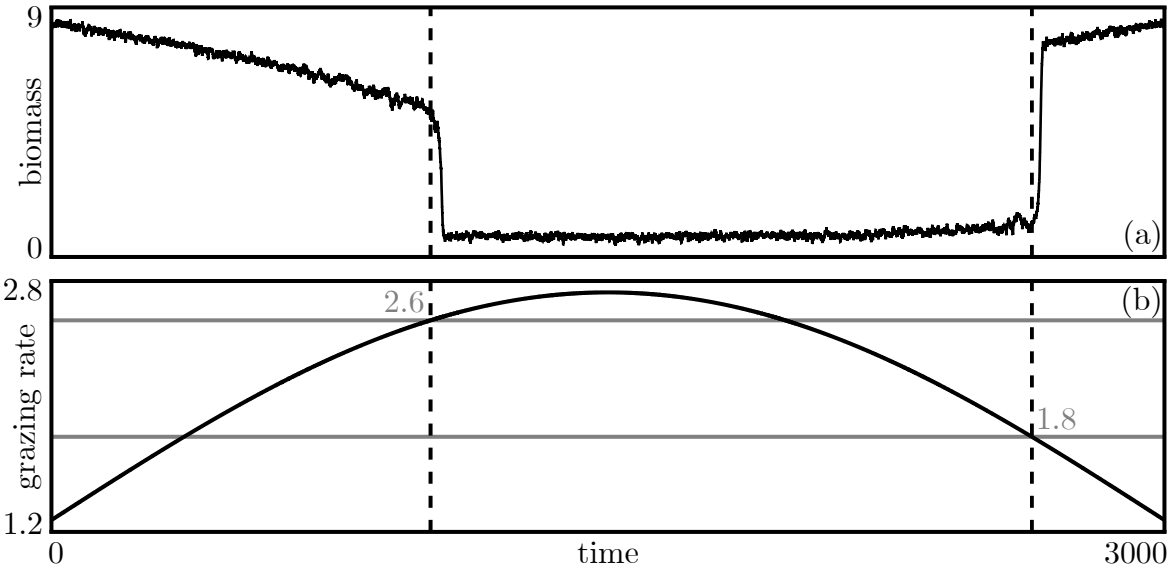


Figure 1.1: A simulation of May's model of ecosystem collapse and recovery: panel (a) shows the system's state as the level of biomass present at a given time; panel (b) shows the input forcing to the system as the grazing rate, also changing over time. Notice how increasingly larger levels of grazing slowly degradate the vegetation biomass in the system until a sudden collapse to a state of almost no cover once the grazing reaches an approximate value of 2.6, indicated by the leftmost dashed black line. The response of the system however does not immediately react in the opposite direction when the grazing rate is brought back to a value below the critical one. Instead, the vegetation cover remains in a state of collapse until the forcing is decreased all the way to approximately 1.8 (corresponding to an approximate 30% reduction from the critical value) in order for the system to recover the previous, healthy state.

new, unsustainable state is also stable however, meaning that the ecosystem will remain in its collapsed state unless a restoring action is enforced. Interestingly however, simply driving the grazing rate back to pre–breakpoint levels may not suffice to guarantee the reversal of the unhealthy state following the critical transition (as shown in Figure 1.1). This increase in the restoring effort can sometimes be so significant that the tipping point may be de–facto irreversible, leaving the ecosystem in a permanent collapsed state.

Despite the rather severe disruptions caused by such events, critical transitions appear to be ubiquitous across many human and natural systems, with varying degrees of complexity and of substantially different nature. Examples of tipping events leading to catastrophic phenomena can be found in extreme paleoclimatic events [49, 5], ecosystems degradation [58, 48], spikes in neurobiological activity [10], onset of asthma and ventilation defects [68, 33], anthropogenic shut–down of ocean currents

[8, 27], hysteresis in laser systems [40], instabilities in magnetically confined plasma [22] and thermoacustic devices [60], capsizing of marine vessels [66], formation of traffic congenstion [17], financial crashes [67, 45] and civilization collapse [64, 31].

1.1 Historical context

Perhaps unsurprisingly, shortly after May's paper on ecosystems collapse, several attempts in characterizing these catastrophic events in a systematic manner followed. In 1984 Wissel observed that the return time of perturbations towards the stable steady–state increases the closer the system is to the breakpoint [71]. Two years later Wiesenfeld and McNamara realized that, more generically, systems approching bifurcations display amplifications of the spectral properties of their signal due to weakening in the attraction of the favourable steady–state [70]. This *"universal law"* was linked to the concept of resilience in (stochastic) ecological systems by Ives in 1995 which identified an increase in variability of population densities relative to increases in environmental degradations [41].

This characteristic phenomenon, later named *critical slowing down* (CSD) in 1998 [35], usually refers to the lethargic decay of stochastic fluctuations in the neighbour of a local attractor as it approaches a tipping point. In particular, the time for these perturbations to relax back to the stable equilibrium is exponential for an attractor that is far enough from the breakpoint, and algebraic, for the same attractor, close to the breakpoint. The onset of CSD prior to a critical transition has since been heavily exploited by several communities to attempt to forecast catastrophic events before those points of no return.

First, in 2001 Scheffer et al. proposed a model of stochastic resilience in ecosystems in which a gradually changing condition, such as the inflow of toxic chemicals and nutrients, or the loss of species diversity, can be perturbed by seemingly random events such as weather extremes, wild fires or pests outbreaks [63]. The resulting model, corroborated by empirical evidence in lakes and oceans ecosystems, was the first attempt at combining multi–stable, deterministic regimes of long–term behaviour with short–term, stochastic fluctuations in the state of a complex system. This was followed by a similar modeling choice in 2003 by Kleinen and Held, which once again used a stochastic approach to detect treshold switchings in the North–Atlantic thermohaline circulation within both conceptual [44] and intermediate complexity models [39]. Finally, in 2006 Carpenter and Brock used timeseries simulations of a lake model

to forecast its eutrophication, with several decades in advance, by detecting CSD via increases in the signals' standard deviation [13].

These early works in the first half of the 2000s, spurred a significant uptake in the prediction of catastrophic shifts in the stable regimes of ecological and climate systems for the remaining part of the decade; here we cite but a few [50, 57, 15, 25, 37, 6, 20, 37, 12, 14, 24, 34]. By the early 2010s the research question of detecting CSD to forecast tipping points in natural systems had spread quite substantially across many scientific communities up to the point that both early reviews [62, 65] and criticisms [38, 29] started to appear.

1.2 A mathematical framework

All of the early endevours in the prediction of critical transitions so far, while relying on the mathematical characterization of slower return times provided by CSD, did not rigorously quantify the proposed statistical indicators as signals of early–warning. In other words, a lack of mathematical justification, beside few qualitative arguments made thus far, started to, justifiably, raise some concerns and skepticism regarding the range of applicability of this new theory [9, 7, 43, 11].

In 2011, Christian Kuehn proposed a framework based on slow–fast stochastic differential equations (SDEs) for the characterization of critical transitions in complex system and the statistical detection of CSD from their timeseries [46]. The starting point is to consider the following SDE

$$dX_t = f(X_t; \mu)dt + \sigma dW_t, \tag{1.1}$$

where $X_t \in \mathbb{R}^d$ is a collection of $d \geq 1$ random variables representing the state of the system (i.e. its response) at any given time t, $\mu \in \mathbb{R}^p$ is a $p-$dimensional parameter vector representing the external forcing acting on the system (i.e. its input), W_t is a canonical Wiener process of standard increments (i.e. $W_{t+s} - W_t \sim \mathcal{N}(0, s)$), $f : \mathbb{R}^d \to \mathbb{R}^d$ is a sufficiently smooth vector field quantifying the (deterministic) drift of the state variables and $\sigma > 0$ quantifies the magntitude of the noisy fluctuations induced by the diffusion process W_t relative to the drift.

Steady–states of a complex system correspond to equilibria of the drift term in (1.1), that is points $x_* \in \mathbb{R}^d$ in the state space for which the vector field vanishes, i.e. $f(x_*) = 0$. Observable steady–states, that is states for which the complex system is measured in the real–world, are further characterized by the spectral properties of the

linearized vector field f. In particular, an equilibrium $x_+ \in \mathbb{R}^d$ is (linearly) stable if the entirety of the spectrum of the Jacobian of f, evaluated at x_+, is in the complex half–plane with negative real part, i.e.

$$x_+ \quad \text{is stable} \quad \iff \quad \mathfrak{Re}\Big[\mathrm{Spec}\big(Df(x_+)\big)\Big] < 0. \tag{1.2}$$

A stable equilibrium attracts nearby regions of the state space which means that (relatively small) stochastic fluctuations around x_+ will robustly return back to it. Conversely, the Jacobian of f at an (linearly) unstable equilibrium $x_- \in \mathbb{R}^d$ features a spectrum entirely contained in the positive real half of the complex plane, i.e.

$$x_- \quad \text{is unstable} \quad \iff \quad \mathfrak{Re}\Big[\mathrm{Spec}\big(Df(x_-)\big)\Big] > 0. \tag{1.3}$$

To produce critical transitions using (1.1), one must allow for either the existence, stability or both of a stable equilibrium x_+ to be parametrized by the external forcing μ. A prototypical example under this framework is the saddle–node normal form for which $f(x;\mu) = -\mu - x^2$. It is easy to deduce that for every strictly negative value of μ there exist two distinct branches of equilibria of f, namely $x_\pm(\mu) = \pm\sqrt{-\mu}$; the positive branch $x_+(\mu)$ has a negative Jacobian $Df(x_+(\mu)) < 0$, which makes it a stable equilibrium, while the negative branch $x_-(\mu)$ has a positive Jacobian $Df(x_-(\mu)) > 0$, which makes it an unstable equilibrium. At $\mu = 0$ these two branches collide and collapse into a unique equilibrium at the origin $x_+(0) = x_-(0) = 0$ which results in a saddle–node (often called *fold*) bifurcation. Finally, for any strictly positive value of μ no real–valued root of f exists and as such no equilibria are left in the state space $\mathbb{R}$; the system can thus never settle on a steady–state and its fate is to *escape* at infinity (finite time blow-up).

Kuehn noted that, by making the parameter itself time–dependent, one can drive the state of the system across the qualitative change prescribed by the saddle–node bifurcation. Specifically, by setting the deterministic part of (1.1) to be a slow–fast (nonautonomous) drift of the form

$$\begin{aligned} dX_t &= f(X_t, \mu)dt + \sigma dW_t, \\ d\mu &= \varepsilon dt, \end{aligned} \tag{1.4}$$

where $0 < \varepsilon \ll 1$ denotes the timescale separation between the evolution of X_t and μ, an initially stable, observed state $x(\mu_0)$, $\mu_0 < 0$, will inevitably pass through a region

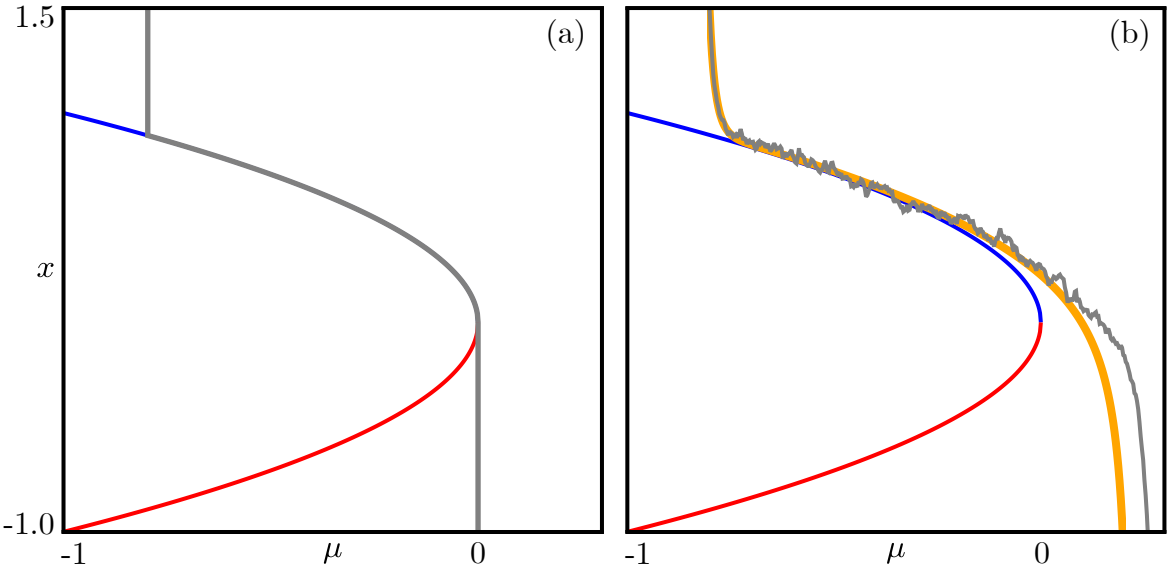


Figure 1.2: Trajectories of a slow–fast system of the form (1.4) associated to a saddle–node normal form $f(x,\mu) = -\mu - x^2$. Panel (a): in the singular limit $\varepsilon \to 0$ a deterministic solution, corresponding to the noiseless case $\sigma = 0$ (grey curve), is decomposed in fast (double arrows) and slow (single arrow) flows; notice that the slow portion of the flow approximates the stable branch $x_+(\mu) = \sqrt{-\mu}$ of the equilibria of f (blue curve). Panel (b): in the nonautonomous regime $\varepsilon = 10^{-1}$ a deterministic solution (orange curve) converges fast to a neighbour of the stable branch and slowly tracks it up to the saddle–node bifurcation after which it escapes at $-\infty$. A sample path associated to a noise level $\sigma = 10^{-2}$ (grey timeseries) follows the same trend of the deterministic solution despite the stochastic fluctuations.

of instability at $\mu = 0$ and eventually be forced to tip to $-\infty$, thereby resulting in a bifurcation–induced critical transition (see Figure 1.2).

To substantiate the statistical detection of CSD from timeseries however, one needs to consider the statistical properties of noisy sample paths rather than just following its deterministic drift. Kuehn thus considered the singular limit $\varepsilon \downarrow 0$ of (1.1) for which a trajectory X_t evolves around the stable equilibrium $x_+(\mu)$, $\mu < 0$ and its fluctuations attain a Boltzmann–like (stationary) probability distribution of the form

$$p(x) \overset{d}{\approx} N \exp\left(\frac{2}{\sigma^2}\int_I f(x;\mu)dx\right), \tag{1.5}$$

with $N > 0$ being a normalization constant of p over a compact set I that contains $x_+(\mu)$. The existence of a stationary (sometimes referred to as *equilibrium*) distribution (1.5) enables the analytical approximation of the central variance of the random

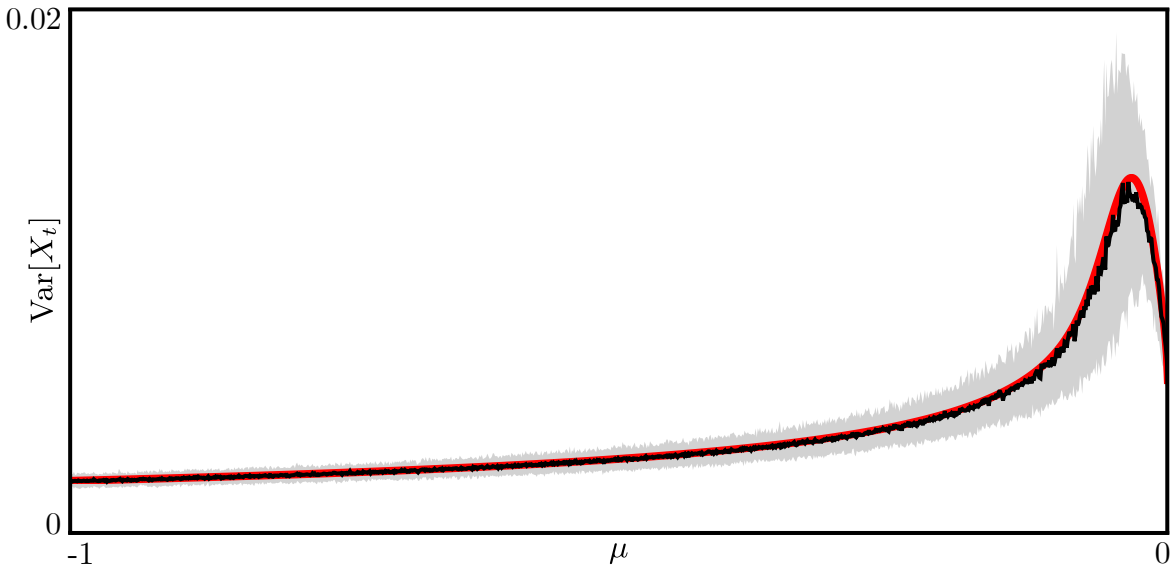


Figure 1.3: Comparison between the variance (1.6) of the equilibrium distribution (1.5) for the saddle–node normal form $f(x;\mu) = -\mu - x^2$ (red curve) and the sample variance of Monte Carlo simulations. At specified parameter values in $[-1, 0)$ an ensemble of 1,000 solutions is computed; for each of these trajectories the sample variance is computed, thus yielding a set of 1,000 measurements. The ensemble mean (black curve) and the $5^{\text{th}} - 95^{\text{th}}$ percentile (gray shaded region) are then extracted.

variable X_t by exploiting its definition

$$\text{Var}[X_t] = \int_I \big(x - \mathbb{E}[x]\big)^2 p(x) dx. \tag{1.6}$$

By considering the case of a saddle–node normal form $f(x;\mu) = -\mu - x^2$, $I = [x_-(\mu), +\infty)$, and plugging (1.5) in (1.6) one can approximate $\text{Var}[X_t]$ by quadrature and thus, in principle, show that an increase in variance of the sample paths does indeed preceed the tipping point. This result is then corroborated by Monte Carlo simulations (see Figure 1.3) in which the variance (1.6) of the equilibrium distribution (1.5) matches the mean sample variance of the solutions of (1.1) with good accuracy.

1.3 Modern developments

In 2012, Ashwin et al. discovered another mechanism that is responsible for generating critical transitions, which is inherently tied to the nonautonomous nature of (1.4). They discovered that when the rate of change ε of the forcing parameter μ is fast enough it would cause the quasi steady–state of a system to escape the basin of

attraction of a stable equilibrium; they named this phenomenon rate–induced tipping [4]. This new phenomon fundamentally differs from the bifurcation–induced critical transitions formalized by Kuehn since rate–induced escapes need not the system to be driven through a local (or global) bifurcation in order for tipping to occur, as later shown in [3, 69]. Ritchie and Sieber later proved that CSD can still be detected for rate–induced tipping via increases in timeseries variance albeit with delay [61]. Rate–induced tipping have since become an active field of research in the modeling of critical transitions, with notable applications recently discovered in ecosystems [58, 48] and ocean currents [1, 51, 16].

Concurrently with these developments, other communities started to research possibilities to extend the theory of CSD–based early–warning signals of tipping points to high–dimensional systems. Specifically, attempts to generalize statistical precursors of critical transitions for spatially–extended systems started to appear in the second half of the 2010s. Early efforts by Kuehn and others on the loss of stability of travelling wave fronts [47] and homogeneous steady–states [36] led the way to later advances in the derivation of early–warning signals for spatial networks which carried over in the early 2020s. Examples of these can be found in [42, 19, 32, 26, 56]. Finally, due to the limitations posed by the collection of large data from spatially extended systems, a question remained on how to optimally observe high–dimensional systems in order to detect CSD. Most recently, works by Lohmann and Gottwald in 2025 [52], and Clarke et al. [21] and [54] in 2026, have started laying the foundations on how low–dimensional observables (i.e. smooth transformations of the state variables) can be extracted from high–dimensional systems in order to anticipate critical transitions through early–warning signals.

1.4 Practical limitations

The mathematical theory of early–warning signals has developed substantially from its early steps, with significant advances been made in the second half of the 2010s and early part of the 2020s. Despite the strong theoretical underpinning provided by the theory, the applicability of these theoretical tools to real–world phenomena has not progressed at the same pace. In most of the above cited works in fact, the detection of CSD via temporal, spatial or spatio–temporal signatures happens almost exclusively in abstract settings, satisfying assumptions that are not necessarily hold when the same techniques are applied data collected from measurements or experiments, rather than simulated.

In 2015, Dakos et al. noted that the application of CSD detectors to real–world ecosystems is limited by *conceptual*, *operational* and *methodological* constraints [23]. Conceptual limitations arise whenever the abrupt regime changes observed in nature are not due to bifurcation–induced critical transitions but instead due to other mechanisms such as noise– or rate–induced escapes; operationally, the early detection of CSD is only reliable whenever there exist sufficient separation in the temporal and spatial scales governing the evolution of the state of the ecosystem and its external forcing, which is not guaranteed to be true universally [59]; in terms of the methodology itself, the observation protocol with which the data is collected highly impacts the significance and sensitivity of these statistical indicators which thus calls for high quality and large quantity of measurements, which may not be satified due to costs and accessibility constraints [18].

A decade later, Ashwin et al. [2] showed how nonlinear and coupling effects in the dynamics of these complex systems heavily affect the predictive power of early–warning signals (named *early–warning skill*) to detect CSD early enough. They observed that early–warning signals with low skill lead to unreliable predictions in ensemble simulations where the rate of missed allarms (i.e. *false negatives*) and spurious warnings (i.e. *false positives*) severely increase. This was recently corroborated in [55] and [30] where high–dimensionality and correlations in the noise structure has been shown to altern the significance of the early–warning signals extracted from observables of these systems.

1.5 Research aim and outline

The above limitations constitute a significant gap between the mathematical abstraction provided by the theory of CSD–based early–warning signals and the real limitations faced by scientists and modelers that seek to apply these methodologies to their data. In this thesis we focused in bridging this gap by providing a more pragmatic approach on the detection of CSD by the latter community. We leveraged mathematical structures in stochastic and nonautonomous dynamics to build end–to–end data pipelines that can be applied out of the box to provide direct, interpretable and actionable policies to prevent catastrophic collapses in complex systems. The research in this thesis address two major themes regarding the limitations of applicability of the theory of early–warning signals by scientists, which we briefly outline below.

Lack of universal interpretability for different systems

At its core, an early–warning signal is a, often statistical, quantity extracted from a monitored observable that somehow provides insights in the proximity of the underlying system to a catastrophic shift in the observed equilibrium. As such, in order for such signals to be applicable, one should be able to interpret such proximity from numerical values attained by the proposed early–warning with reasonable confidence. Rather surprisingly however, the most popular and established detectors of CSD, e.g. the increase in timeseries variance, do not satisfy this requirement. As described in [23] in fact *"one–time measurements of CSD indicators have limited forecasting value"* and only *"once a baseline indicator values is established, then a consistent deviating trend from the baseline could indicate whether a transition is lurking"*. This limitation stems from the fact that traditional CSD–based precursors are not bounded and their numerical value alone does not trivially inform the proximity of a tipping point, rather their trend does. On top of this, systems of different nature, or even the same system under different temporal and spatial scales, do not agree on similar values of variability across time and space, thereby making this limitation a universal feature of the currently used early–warning signals.

To remedy this constraint, we formulate an interpretable early–warning signal that is bounded in $[0, 1]$ for any complex system whose tipping is bifurcation–induced. In particular, values of the proposed quantity close to 0 indicate small likelihood of tipping events to occur in the near future, whereas values close to 1 indicate immediate danger of tipping events to occur in the near future. The proposed early–warning signal is universal in the sense that any system who is slowly driven through a saddle–node tipping, in the presence or relatively small perturbations, will monotonically increase from 0 to 1, regardless of its nature and its temporal or spatial scales.

Lack of optimal observation protocols for extrapolation

Extrapolation methods to estimate the time of tipping of saddle–node types of critical transitions have been proposed for the anticipation of the collapse of the Atlantic Meridional Overturning Circulation by Boers in 2021 [8] and Ditlevsen and Ditlevsen in 2023 [28]. More recently in 2025, Ashwin et al. used the same setup to establish the early–warning skill (or lack thereof) carried by samples of cascading tipping systems [2]. These methods, based on the estimation of the *return* (or *restoring*) *rate* from timeseries data, rely on a set of assumptions that the data must specify in both quality

and quantity. Furthermore, these estimators are negatively biased due to finiteness and non–stationarity of the samples, which previous works do not account for.

In attempt to overcome the limitations provided by these statistical estimations, we build an alternative pipeline that relaxes some of these assumptions. On top of this, we derive optimal conditions for the collection of the data that makes the extrapolation problem feasable and more accurate. The resulting observation protocol thus provides strict and interpretable conditions for scientists and modellers to directly apply forecasting techniques to their timeseries, while at the same time ensuring the minimization of biases and the quantification of uncertainties associated to the statistical estimation.

Outline

The thesis is organized as follows.

In **Chapter 2** we provide the necessary background for the mathematical grounding of critical transitions in slow–fast SDEs, including a review of fundamental results in stochastic dynamics, statistical estimation, nonautonomous differential equations and large deviations.

In **Chapter 4** we tackle the issue of lack of universal interpretability by deriving a new early–warning signal that systematically increases from 0 to 1 as a bifurcation–induced tipping is approached. In addition of this derivation we construct a computational algorithm that estimates the statistical quantity based on input timeseries that does not require an underlying knowledge of the physics of the system.

In **Chapter 3** we address the problem of forecasting tipping times via extrapolation by deriving the necessary conditions for an optimal observation protocol to be employed by the end–users. We also show how reduction in bias and quantification in the uncertanty of these estimators can be achieved by considering more general cases of quasi–stationary samples.

In **Chapter 5** we show how both methodologies can be combined to accurately predict tipping in real–world, high–dimensional timeseries with high degree of confidence.

Finally, in **Chapter 6** we summarize the main results discussed throughout the thesis and indicate possible future directions for the research to develop.

Chapter 2

Background

Emphasize that not all critical transitions are predictable under the currently developed framework and cite criticism within the community. Stress how the current theory is largely developed for stochastically forced systems that experience a slow passage through a saddle-node bifurcation. The goal of this section is to select a candidate timeseries from the real world (given in Motivation) and fit a saddle-node model with stochastic fluctuations. To do so we need two main ingredients: geometric singular perturbation theory (or Fenichel theory) and large deviation theory (or Freidlin and Wentzell theory).

2.1 Nonautonomous dynamics

Introduce the autonomous saddle-node normal form and an example of coordinate transformations for the saddle-node bifurcation theorem. Extend it to the dynamic case with the notion of pullback attractors.

2.2 Stochastic calculus

Introduce the fundamentals of stochastic calculus including the OUP and the FPE.

2.3 Large deviations in slow-fast systems

State that the correct mathematical interpretation of critical transitions is that of rare events in small perturbations of dynamic saddle-node systems. Introduce the two ingredients separately.

2.3.1 Geometric singular perturbation theory

Main results of Fenichel theory.

2.3.2 Large deviation theory

Main results of Freidlin-Wentzell theory.

Chapter 3

A data–driven forecasting pipeline

Chapter 4

An interpretable early-warning signal

Chapter 5

Applications

Chapter 6

Conclusions

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